Jinsong Liu

Contact Information

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Research Summary

My research develops reinforcement learning methods for agentic reasoning and tool-use in LLM-based systems, grounded in theoretical work on value iteration, policy optimization, and reward preference modeling. I design agents that reason over multi-step tasks, use tools autonomously, and learn from human and AI feedback, with applications in healthcare AI at Weill Cornell Medicine and GPU-accelerated large-scale optimization. This work has produced open-source releases widely adopted by the research community (**1,500+** and **200+** GitHub stars).

Academic Background

Cornell University
Postdoctoral Associate, Weill Cornell Medicine — Jun. 2025 – Current
Research direction: AI for Healthcare

Cornell University
Visiting PhD Student, SC Johnson College of Business — Jul. 2024 – May. 2025
Research direction: LLM Agent for Decision Making

Shanghai University of Finance and Economics
Ph.D. in Management Science and Engineering — Sep. 2020 – May. 2025
Thesis: *Efficient and Adaptive Reinforcement Learning: Explorations and Improvements in Value Iteration, Policy Optimization, and Reward Preference Modeling*

Tianjin University
B.E. in Electronic Engineering — Aug. 2016 – Jun. 2020
B.S. in Computer Science (Double Degree)

Publications

†: equal contribution, ($\alpha\beta$): alphabetical order, *: corresponding author.

Working Papers & Preprints

WP1. **Liu, J.**, Jiang, Y., Krishnan, R., Padman, R., Zhang, Y., & Bian, J. (2026). Closing Reasoning Gaps in Clinical Agents with Differential Reasoning Learning. arXiv preprint arXiv:2602.09945.

WP2. **Liu, J.**, Brown, K., Ancker, J. S., RoyChoudhury, A., Moliere, N., Wright, A., Rosen, A., Han, J. H., Steel, P. A. D., Zhang, Y. (2025). Multi-Site Validation of Large Language Models to predict Emergency Department Return Visit With Admission. *In preparation for Nature Communications.*

WP3. Wan, Y., Wang, J., Li, L., **Liu, J.**, Zhu, R., & Zhu, Z. (2025). PokeeResearch: Effective Deep Research via Reinforcement Learning from AI Feedback and Robust Reasoning Scaffold. arXiv preprint arXiv:2510.15862.

WP4. Liang, K., Zhang, C., **Liu, J.***, Ge, D., & Wang, Z. (2025). A Column-Generation-Based Framework for Dynamic Network Optimization with Service Customization. Available at SSRN 5391463.

WP5. **Liu, J.**, Ge, D., & Zhu, R. (2024). Reward learning from preference with ties. arXiv preprint arXiv:2410.05328.

WP6. Qin, H., Zhao, X., **Liu, J.***, Ge, D., & Zhu, R. (2024). Dispatching Automated Guided Vehicles Using Efficient Data-Driven Optimization. Available at SSRN 4959037.

WP7. Lu, H., Yang, J., Hu, H., Huangfu, Q., **Liu, J.**, Liu, T., ... & Ge, D. (2023). cupdlp-c: A strengthened implementation of cupdlp for linear programming by c language. arXiv preprint arXiv:2312.14832.

WP8. **Liu, J.**†, Xie, C.†, Deng, Q., Ge, D., & Ye, Y. (2023). Stochastic Dimension-reduced Second-order Methods for Policy Optimization. arXiv preprint arXiv:2301.12174.

Journal Publications

J1. ($\alpha\beta$) Deng, Q., Feng, Q., Gao, W., Ge, D., Jiang, B., Jiang, Y., ... & Zhang, C. (2025). An enhanced alternating direction method of multipliers-based interior point method for linear and conic optimization. INFORMS Journal on Computing, 37(2), 338-359.

J2. ($\alpha\beta$) Ge, D., **Liu, J.**, Liu, T., Tan, J., & Ye, Y. (2024). Algorithm 1053: SOLNP+: a derivative-free solver for constrained nonlinear optimization. ACM Transactions on Mathematical Software, 50(4), 1-24.

Conference Publications

C1. **Liu, J.**, Xie, C., Deng, Q., Ge, D., & Ye, Y. (2024, March). Sketched newton value iteration for large-scale Markov decision processes. In Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 38, No. 12, pp. 13936-13944).

Open-Source Software

- **PokeeResearch-7B Agent (1,500+ stars)**: State-of-the-art 7B DeepResearch agent that leverages web search and content reading capabilities to answer complex questions using the most up-to-date information available online.
  Repository: github.com/Pokee-AI/PokeeResearchOSS
- **cuPDLP-C (200+ stars)**: Industry's first GPU-accelerated primal-dual LP solver implemented in C/CUDA, achieving significant speedup over CPU baselines on large-scale instances.
  Repository: github.com/COPT-Public/cuPDLP-C
- **ABIP**: An implementation of an ADMM-based interior-point method for linear and conic optimization problems.
  Repository: github.com/leavesgrp/ABIP

Professional Experience

Pokee AI
Algorithm Research Scientist (Advisor) Jan. 2025 – May. 2025
Remote

- Designed and implemented agentic RL algorithms for LLM-based tool-use agents, advancing capabilities in autonomous multi-step reasoning and decision-making

- Co-architected DeepResearch agents achieving SOTA performance among similar-scale open-source models through novel RL techniques and reward modeling strategies
- **1,500+ GitHub stars** on the open-sourced implementation (PokeeResearchOSS), demonstrating broad research community adoption

Cardinal Operations (Shanshu)
Algorithm Research Intern — Apr. 2023 – Apr. 2025
Shanghai, China

- Bridged machine learning and mathematical programming by integrating deep reinforcement learning into large-scale optimization, enabling learned decision-making at production scale
- **First GPU-accelerated** LP solver in industry: built cuPDLP-C (**200+ stars**) in C/CUDA, solving problems with millions of variables—GPU systems expertise directly transferable to large-scale agent training

JD.com
Algorithm Research Intern — Dec. 2021 – Aug. 2022
Beijing, China

- **100+ AGVs** coordinated in real time via multi-agent algorithms for automated warehouse systems, optimizing dispatching under dynamic constraints
- **90% reduction** in task incompletion rate through a deployed learning-based scheduling framework, directly impacting production operations

Conference Talks & Workshops

- *Multi-Site Validation of Large Language Models to predict Emergency Department Return Visit With Admission*
  AMIA Annual Symposium, Atlanta, USA, November 2025.
- *LLMs for Choice Prediction: A Generalized Direct Preference Optimization Approach*
  INFORMS Annual Meeting, Atlanta, USA, October 2025.
- *LLM-Based Reasoning for Return Visit Admissions in the Emergency Department*
  INFORMS Annual Meeting, Atlanta, USA, October 2025.

Teaching Experience

Guest Lecturer, Operations Research for Healthcare — Mar. 2026
Institution: Weill Cornell Medicine

- Taught optimization, simulation, and decision analysis methods applied to healthcare; covered LP/IP modeling for clinical resource allocation and patient outcomes.

Guest Lecturer, AI for Healthcare — Nov. 2025
Institution: Weill Cornell Medicine

- Taught LLM foundations and healthcare data preprocessing for graduate students; demonstrated clinical applications through case studies.

Skills

Programming: Python, CUDA, Julia, C\C++, Matlab, SQL

ML/AI Frameworks: PyTorch, HuggingFace Transformers, vLLM, LangChain
Optimization/Modeling: Gurobi, COPT, MOSEK, CLP, Cplex